

ZOMATO RESTAURANT EXPANSION

ABSTRACT

This project showcases a detailed restaurant analytics dashboard created with Google Sheets to aid in pinpointing optimal areas for expanding a restaurant business. The objective is to facilitate informed decision-making by examining crucial factors such as the presence of restaurants, customer reviews, pricing structures, and regional competition across various nations.

The dataset employed encompasses comprehensive details on thousands of restaurants, including their geographical locations, types of cuisine, pricing strategies, delivery and reservation options, customer feedback, and establishment dates. The information was meticulously cleaned and structured to ensure precise analysis.

The dashboard delivers visual representations through comparisons by country and year, assisting management in comprehending areas with high restaurant density and identifying potential opportunities. It also showcases cities and nations that present favourable conditions for new openings, characterized by fewer competitors, strong ratings, and reasonable pricing.

Further analysis involves assessing the popularity of different cuisines, tracking customer feedback trends, and recognizing leading competitors in distinct regions. The report also provides recommendations on optimal pricing, delivery methods, and booking features that resonate with customer preferences.

By integrating all these insights into a single cohesive dashboard, this project presents a practical, data-driven strategy for planning restaurant expansions in both local and international markets.

Objective Questions

1. What is the total no. of tables present in the data?

Ans: 2

2. What is the total no. of attributes present in the data?

Ans: 22 (20 in Raw Data + 2 in Country_Description)

3. How many categorical columns are there in the data?

Ans: 15 (13 in Raw Data + 2 in Country_Description)

In Raw Data,

Restaurant Name, Country Code, City, Address, Locality, Locality Verbose, Cuisines, Currency, Has_Table_Booking, Has_Online_delivery, Is_delivering_now, switch_to_order_menu, Price Range

In Country Description,

Country Code, Country Name

4. The data consists of some inconsistent and missing values so ensure that the data used for further analysis is cleaned.

Ans:

Step 1: Handling Missing Values

Column: Cuisines

Identified Issues: 9 missing values in cuisines column.

Steps Followed:

I had used search method to search the blank columns and applied filter by colour and also identified that, they belong to Country Code 216.

Analysis for operation:

I had created a pivot table in Objective Sheet for Country Summary

Pivot table 1: Country Summary

Rows: Country Code

Values: RestaurantId (Count Unique), Rating (Average)

Output: Count: 434 and average rating: 4.0

Pivot table 2: Cuisine Vs Country

Rows: Cuisines

Columns: Country Code

Values: Rating (Average)

Output for blank Cuisines: 4.011

Conclusion:

Since, average of blank cuisine analysis and overall average, removing them won't affect analysis.

So, we can delete those 9 rows.

Step 2: Handling Inconsistent Values

Column: Restaurant Id

Issues: Some Restaurant Id values had fewer than 8 digits.

Approach: Using Helper Column, I identified length of A2 cell and highlighted those who are less than that using Conditional Formatting.

Now, I was formatting the column using Custom Number Format with 00000000 (8 Zero's)

Column: Address, Locality and Locality Verbose

Issues: Leading and trailing Spaces.

Approach: Using Helper column, I had compared the length Normally with the trimmed length.

And used the Trim function to remove those.

Step 3: Adding Additional Preprocessing

(i) **Goal:** Adding Country Names

Approach: Using XLOOKUP based on Country Code.

Formula: XLOOKUP(C2,'countrydescription'!\$A\$2:\$A\$16,'countrydescription'!\$B\$2:\$B\$16)

(ii) **Goal:** Adding Formatted Date Column and Extracted Year Column

Approach: Replace Underscores with hyphens and selecting right 4 digits for year.

Formula: DATEVALUE (SUBSTITUTE (AB2, "_", "-")) and RIGHT (AC2, 4).

(iii) **Goal:** Adding Average Cost Of Two INR Conversion

Approach: I was using Google Finance function to obtain the currencies w.r.t INR in Country Description sheet.

And using XLOOKUP function, I moved rates w.r.t INR to Raw_Data table, later multiplied with their respective amounts.

Formula: IF(E2<>\$E\$2, GOOGLEFINANCE("Currency:"&E2&\$E\$2), 1) and XLOOKUP(M141,'countrydescription'!\$C\$2:\$C\$16,'countrydescription'!\$F\$2:\$F\$16)

5. Using the LookUp functions, fill up the countries in the original data using the country code.

Ans: Yes, I added Country Name in Raw_Data in D column, based on country code from C column and used Xlookup function for Country_code and Country_Name in Country_Description Sheet.

Formula:

XLOOKUP(C3,'country_description'!\$A\$2:\$A\$16,'countrydescription'!\$B\$2:\$B\$16)

6. Create a table to represent the number of restaurants opened in each country.

Ans: Here we create a table which is known as Pivot Table using Country_Name (in Rows) and Count of RestaurantIDs in values from Raw_Data.

Country Name	COUNTA of Rest
Canada	4
Qatar	20
Singapore	20
Sri Lanka	20
Indonesia	21
Philippines	22
Australia	24
Turkey	34
New Zealand	40
Brazil	60
South Africa	60
United Arab Emi	60
United Kingdom	80
United States of	434
India	8652
Grand Total	9551

7. Also, the management wants to look at the number of restaurants opened each year, so provide them with something here.

Ans: Here, for getting number of restaurants opened each year, we will create a pivot table using Opening Year in rows and count of Restaurants in values from Raw_Data.

OPENING_YEAR	COUNTA of RestaurantID
2010	1080
2011	1098
2012	1022
2013	1061
2014	1051
2015	1024
2016	1027
2017	1086
2018	1102
Grand Total	9551

Opening Year vs No. of Restaurants

COUNTA of Rest	OPENING_YEAR								
Country Name	2010	2011	2012	2013	2014	2015	2016	2017	2018
Australia	4	1	3	6			4	2	1
Brazil	5	12	2	8	11		9	5	4
Canada		1		1			1		1
India	995	995	911	954	946	918	938	992	1003
Indonesia	1	5		1	4	3	1	1	5
New Zealand	4	6	4	2	4	4	4	5	7
Philippines	6	3	2	1	2	1	1	2	4
Qatar	4	1	2	4	2	2	4		1
Singapore	2	3	4	1	2	2	1	2	3
South Africa	4	4	7	8	5	7	10	9	6
Sri Lanka	1	2	3	4	2	3	2	2	1
Turkey	3	2	1	5	6	4	4	3	6
United Arab Emi	4	3	16	6	9	8	2	6	6
United Kingdom	9	6	12	10	11	5	7	12	8
United States of	38	54	55	50	47	53	46	46	45

Country Vs Opening Year Vs No. of Restaurants

8. What is the total number of restaurants in India in the price range of 4?

Ans: 388

Formula:

COUNTIFS('Raw Data'!D2:D9552, "India", 'Raw Data'!T2:T9552, 4)

9. What is the average number of voters for the restaurants in each country according to the data?

Ans: Here, for getting average number of voters for restaurants in each country, we will create a pivot table using Country_Name in rows and Votes in Values, aggregated by AVERAGE.

Country_Name	AVERAGE of Votes
Australia	111.4166667
Brazil	19.61666667
Canada	103
India	137.212552
Indonesia	772.0952381
New Zealand	243.025
Philippines	407.4090909
Qatar	163.8
Singapore	31.9
South Africa	315.1666667
Sri Lanka	146.45
Turkey	431.4705882
United Arab Emi	493.5166667
United Kingdom	205.4875
United States of	428.2211982
Grand Total	156.9097477

10. Calculate the average rating for all the restaurants that have price_range < 4 and provide online delivery. Use only the “IF” function, Logical Operators, and Aggregation functions to solve this problem.

Ans: 3.2738

Formula: AVERAGEIFS ('Raw Data'!Z2:Z9552, 'Raw Data'!T2:T9552, "<4", 'Raw Data'!Q2:Q9552, "Yes")

11. Using Conditional formatting highlight the rows of restaurants that are located in the countries or cities that you’ve suggested to the management for opening new restaurants.

Ans:

Criteria:

- i. Lesser number of restaurants in a country (<60).
- ii. Higher average ratings in a country (>3.8).
- iii. Above average expenditure of cost of two in a country

Based on these, I had Highlighted the rows in Raw_Data using conditional formatting.

Formula:

=OR(\$D2="Qatar", \$D2="New Zealand", \$D2="Indonesia", \$D2="Sri Lanka")

12. Create a new customized price column that consists of the abbreviation/symbol of the currency along with the Average_cost_for_two value. [Use string operations to do this task]

Ans: Yes, I had created separate column for Customized_Average_cost_for_two, in Raw_Data (X column) using String operations and also for converted_price in INR.

Formula:

=N2& " " &W2 and = "INR"& " " &Y2

13. How can you create an array formula in Excel or Google Sheets to count the number of restaurants listed that do not offer online delivery, are in the lowest price range, and have an average cost for two people less than or equal to 250 Indian Rupees?

Ans: 1676

Let’s Assume, True means 1 and False means 0.

Here, take the mentioned three conditions and mark them as 1 or 0 (True/False), If all conditions are met, then add those, if not leave it.

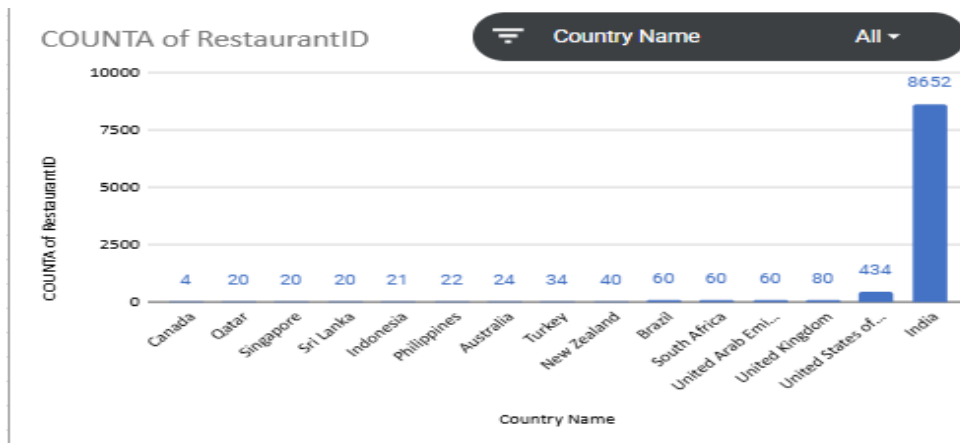
Formula: ARRAYFORMULA(SUM(('Raw Data'!Q2:Q9552="No")*('Raw Data'!T2:T9552=1)*('Raw Data'!Y2:Y9552<=250)))

Subjective Questions

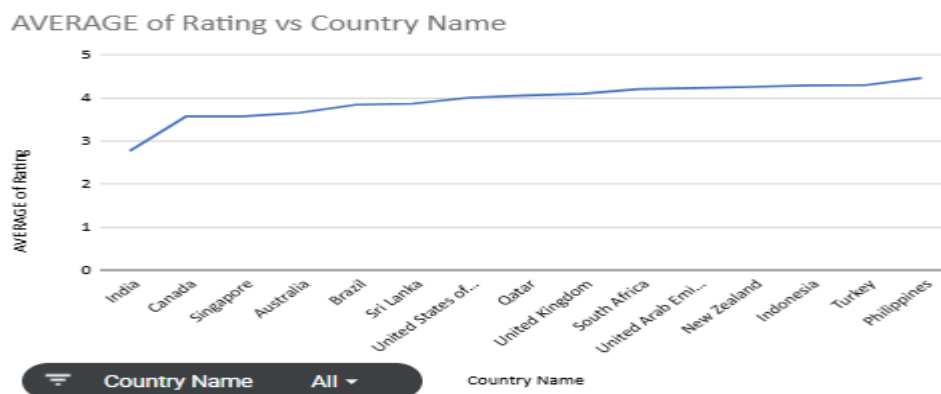
1. Suggest a few countries where the team can open newer restaurants with lesser competition. Which visualization/technique will you use here to justify the suggestions?

Ans: For this first question, I had followed three different approaches for selecting the countries for our restaurant expansion.

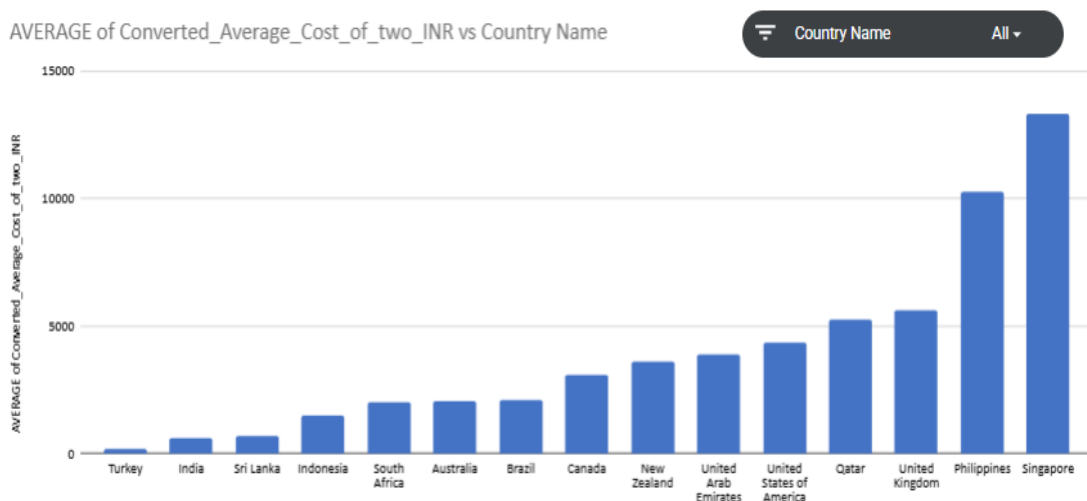
Approach 1: Countries with a smaller number of restaurants gives us a choice



Approach 2: Countries with high average rating gives us an option



Approach 3: Countries with above average expenditure on the average cost of two people



Insights:

From approach 1, we get 10 countries with a smaller number of restaurants:

Canada, Qatar, Singapore, Sri Lanka, Indonesia, Philippines, Australia, Turkey, New Zealand, Brazil

From approach 2, we get 10 countries with above average ratings:

Philippines, Turkey, Indonesia, New Zealand, UAE, South Africa, UK, Qatar, USA, Sri Lanka

From approach 3, we get 10 countries with above average values:

Indonesia, South Africa, Australia, Brazil, Canada, New Zealand, UAE, USA, Qatar, UK

Suggestions:

Qatar, Sri Lanka, Indonesia, New Zealand are suitable countries (common in all 3)

2. Come up with the names of States and cities in the suggested countries suitable for opening restaurants.

Ans: For selecting the cities, I had followed similar approach for countries.

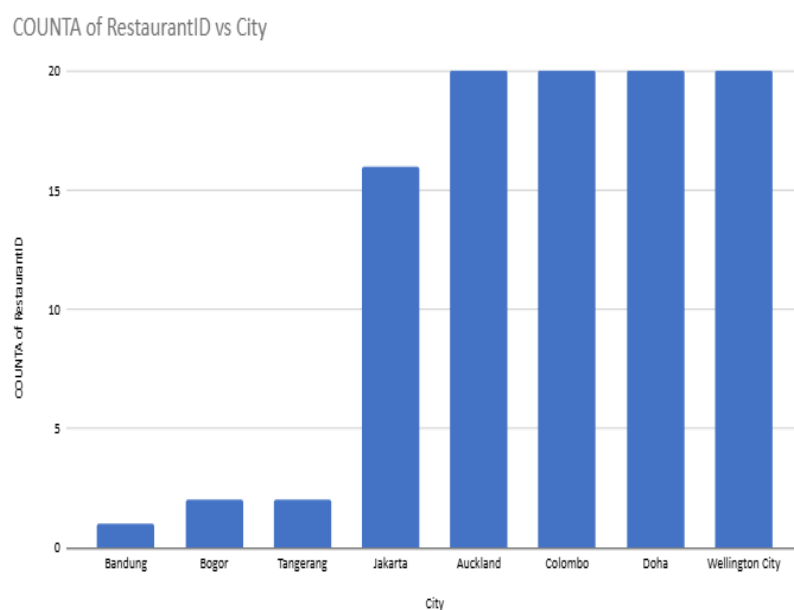
Approach:

Initially, I took the cities with the count of restaurants in it and added the slicer with respect to the Country Names.

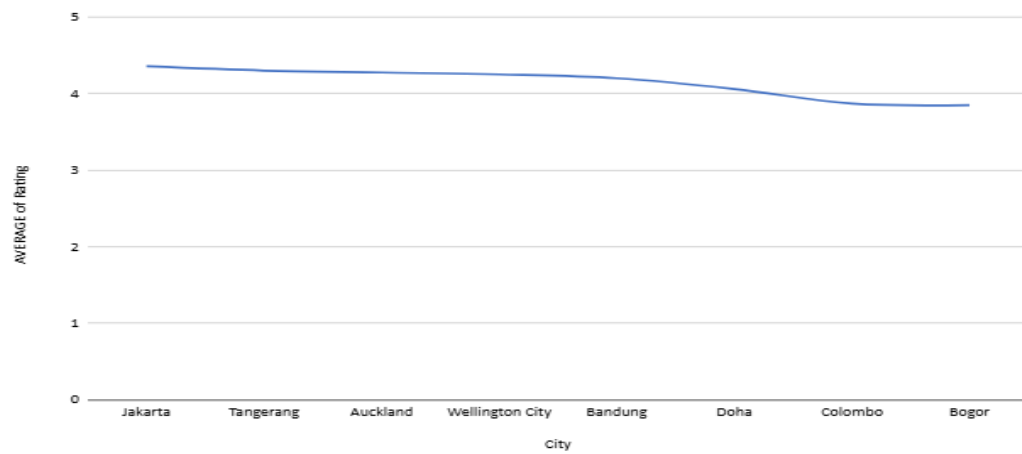
Second, I took the cities with the high average rating and added the slicer with respect to the country names.

Third, I took the cities with above average expenditure of average cost of two people and added the slicer with respect to the country names.

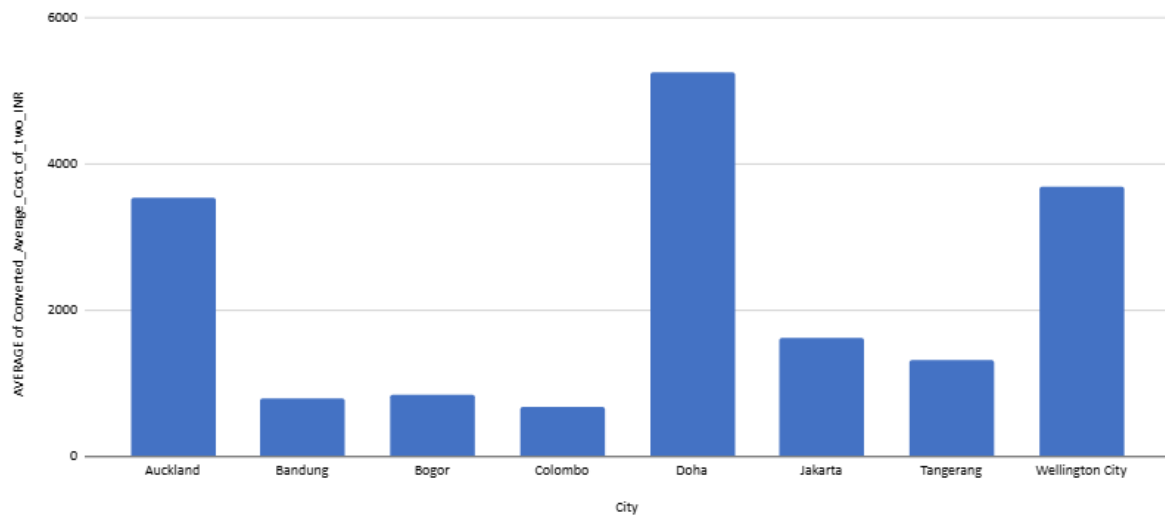
References:



AVERAGE of Rating vs City



AVERAGE of Converted_Average_Cost_of_two_INR vs City



Insights:

For Qatar and Sri Lanka there are only one city. So, we will take them by default

For Indonesia, they are four cities, among 4, Bandung and Bogar stand out due to their average cost of two people.

For New Zealand, they are two cities, among 2, Auckland and Wellington City stand out because they almost have the same values in count of restaurants, average cost of two people.

Suggestions:

In Qatar: Doha

In Sri Lanka: Colombo

In Indonesia: Bandung and Bogar

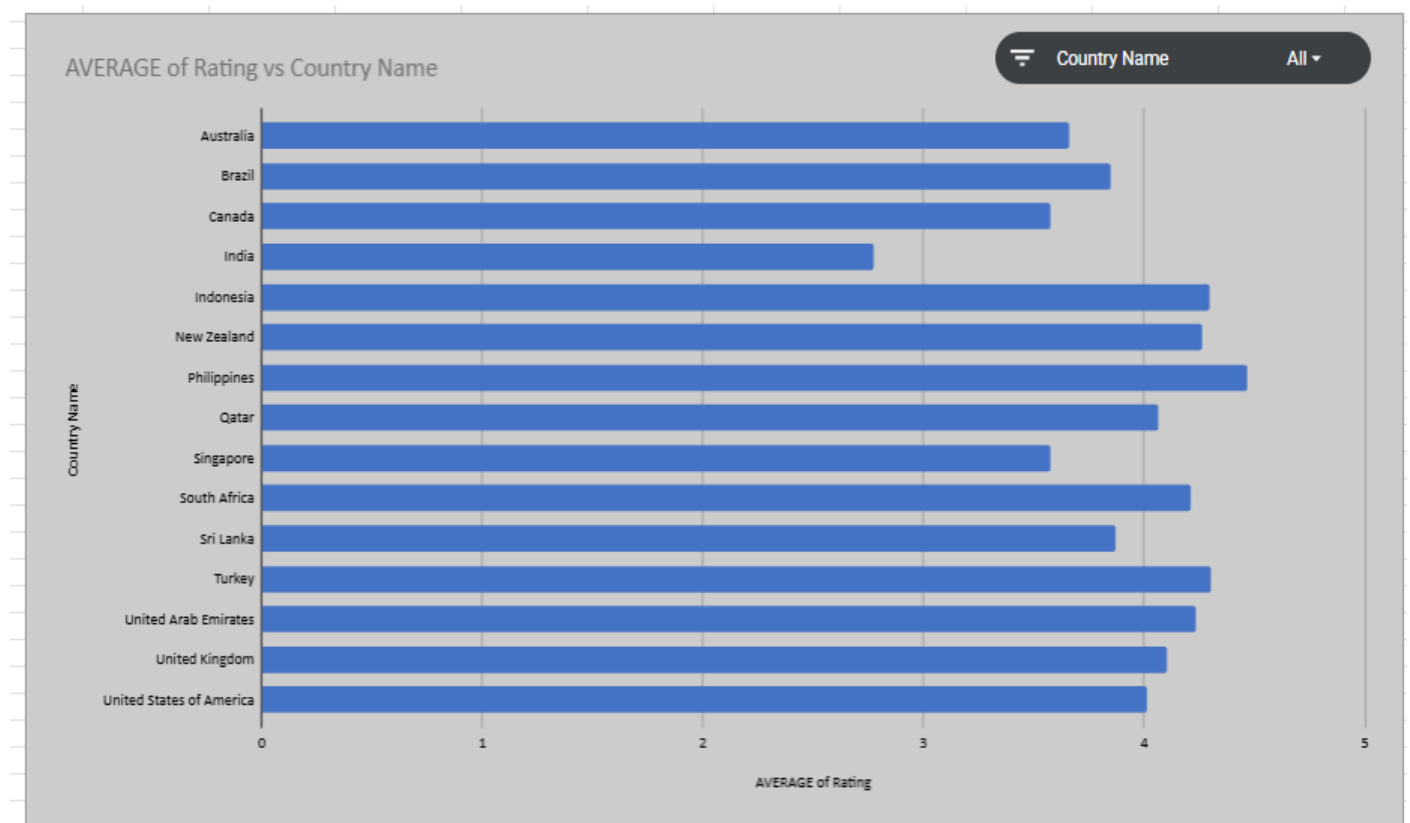
In New Zealand: Auckland and Wellington City

3. According to the countries you suggested, what is the current quality regarding ratings for restaurants that are open there?

Ans: To get the current quality regarding rating for suggested countries, we need two parameters from those respective countries (i) Country Name
(ii) Average Ratings

I had created a pivot table and obtained a column chart using above two parameters.

Country Name	AVERAGE of Rat
Philippines	4.468181818
Turkey	4.3
Indonesia	4.295238095
New Zealand	4.2625
United Arab Emi	4.233333333
South Africa	4.21
United Kingdom	4.1
Qatar	4.06
United States of	4.011290323
Sri Lanka	3.87
Brazil	3.846666667
Australia	3.658333333
Singapore	3.575
Canada	3.575
India	2.770550162



Insights:

Indonesia has high average ratings with 4.29, it shows that, it has high customer satisfaction in its restaurants.

New Zealand follows; it has average ratings of 4.26, it also shows high customer satisfaction in its restaurants.

Qatar has average ratings of 4.06, which also shows its restaurant quality is good.

Sri Lanka has average ratings of 3.87, which explains us that there is a room for potential improvement.

Suggestions:

Indonesia, New Zealand, Qatar has higher quality, we should continue to maintain the higher standards and explore various ways to make customer comfortable.

For Sri Lanka, we should focus on many other things for getting good ratings.

4. Also, what is the current expenditure on food in the suggested countries, so we can keep our financial expenditure in control?

Ans: To Calculate the current Expenditure, we need to find the Average Cost of Two people from each country and convert all currencies to a common one, I was taking INR.

Here, I have taken currency codes for each country from google like INR for Rupee, USD for dollar like that and I had used a formula for Currency Exchange rates and dynamically integrated to IF formula.

Formula:

`IF(E2<>$E$2, GOOGLEFINANCE("Currency:"&E2&$E$2), 1)`

After getting currency Exchange rates I multiplied with Average Cost of two people value and created a new column.

Approach:

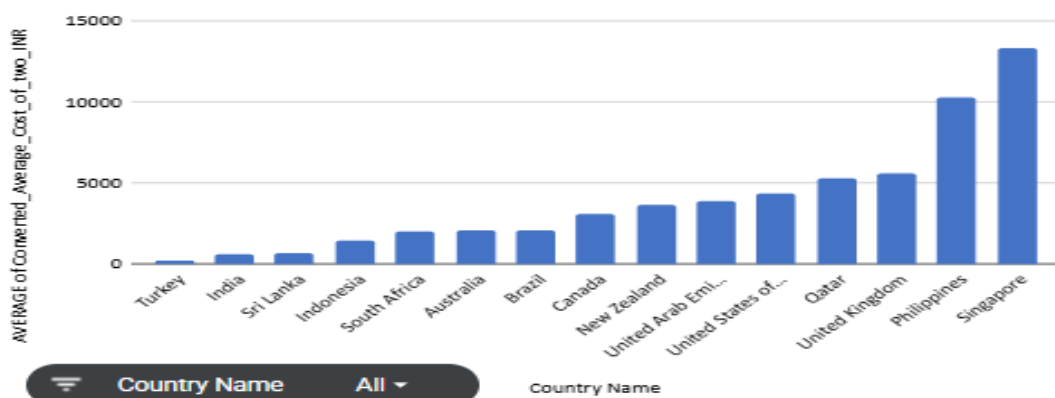
To get the current expenditure on food in the suggested countries, we need two parameters from those respective countries (i) Country Name

(ii) Average of Converted Cost of two people

I had created a pivot table and obtained a column chart using above two parameters.

Country Name	AVERAGE of Cor
Turkey	182
India	625
Sri Lanka	677
Indonesia	1481
South Africa	2006
Australia	2059
Brazil	2097
Canada	3099
New Zealand	3610
United Arab Emi	3874
United States of	4366
Qatar	5254
United Kingdom	5611
Philippines	10278
Singapore	13314

AVERAGE of Converted_Average_Cost_of_two_INR vs Country Name



Insights:

The average expenditure in Qatar is higher than remaining 3 countries with 5254 Indian Rupees. It needs tight budget planning.

New Zealand follows; It is also having higher expenditure than remaining 2 with 3610 Indian Rupees, it needs careful cost control.

Indonesia had moderate and manageable cost with 1481 INR.

Sri Lanka is cost friendly with average costs with 677 INR.

Suggestions:

For Qatar, we can focus on high margin cuisines with small dine-in and a Cloud Kitchen for budget planning.

For New Zealand, we can focus on combo packs and we can charge high and get ingredients from locals for cost cutting.

For Indonesia, we can focus on popular and affordable items.

For Sri Lanka, we can focus on budget friendly dishes.

5. Come up with the names of restaurants from the recommended states that are our biggest competitors and also those that are rated in the lower brackets, i.e. 1-2 or 2-3

Ans: For this, we can use some parameters, so we can get the restaurant names easier.

Approach:

For Big Competitors (For Understanding how they succeed):

Step 1: I had utilized 3 pivot tables to understand the restaurant names, average ratings, price range and their average cost of two people in INR.

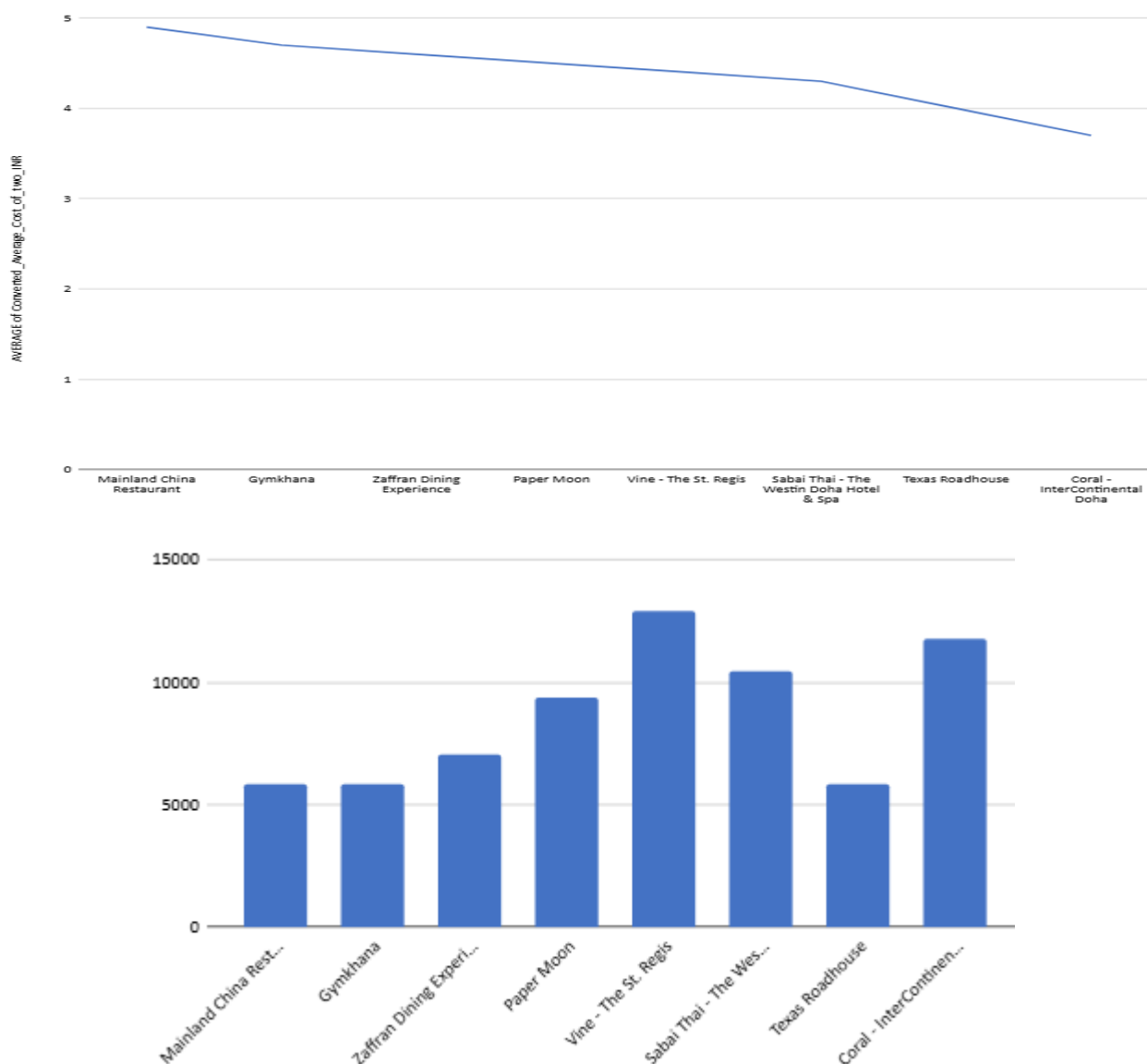
Step 2: For filtering competitors, I was sorting table in decreasing order for high average ratings, selecting 2-4 price range and average cost of two people should be greater than or equal to country average cost of two people.

Step 3: We are analysing restaurants from the suggested countries like Qatar, New Zealand, Indonesia, Sri Lanka.

References:

RestaurantName	AVERAGE of Rating	AVERAGE of Cost	AVERAGE of Price
Mainland China Restaurant	4.9	5870	4
Gymkhana	4.7	5870	4
Zaffran Dining Experience	4.6	7044	4
Paper Moon	4.5	9392	4
Vine - The St. Regis	4.4	12914	4
Sabai Thai - The Westin Doha	4.3	10449	4
Texas Roadhouse	4	5870	4
Coral - InterContinental Doha	3.7	11740	4

AVERAGE of Converted_Average_Cost_of_two_INR vs AVERAGE of Rating



Insights:

In Qatar (Doha):

Mainland China Restaurant with rating 4.9, price range is 4 and Average cost of two is 5870.

Gymkhana with rating 4.7, price range is 4 and Average cost of two is 5870.

In New Zealand (Auckland and Wellington City):

In Auckland,

Depot Eatery and Oyster Bar with rating 4.8, price range is 4 and average cost of two is 4654.

In Wellington City,

Ombra with rating 4.5, price range is 4 and average cost of two is 3620.

In Indonesia (Bandung and Bogor):

Lemon Grass with rating 4, price range 3 and average cost of two is 1316.

In Sri Lanka (Colombo):

Ministry of Crab with rating 4.9, price range is 4 and average cost of two is 1141.

The Sizzle with rating 4.2, price range 3 and average cost of two is 856.

Suggestions:

Understand their cuisines, pricing, service model, quality, ambience etc to analyse which works best at their respective countries and cities.

For Qatar, we can focus on Affordable premium/fusion.

For New Zealand, we can focus on Fusion Indian, strong ambience.

For Indonesia, we can focus on Trendy cuisines with local ingredients.

For Sri Lanka, we can focus on Signature Dishes and branding.

For Weak Competitors (For understanding why they failed):

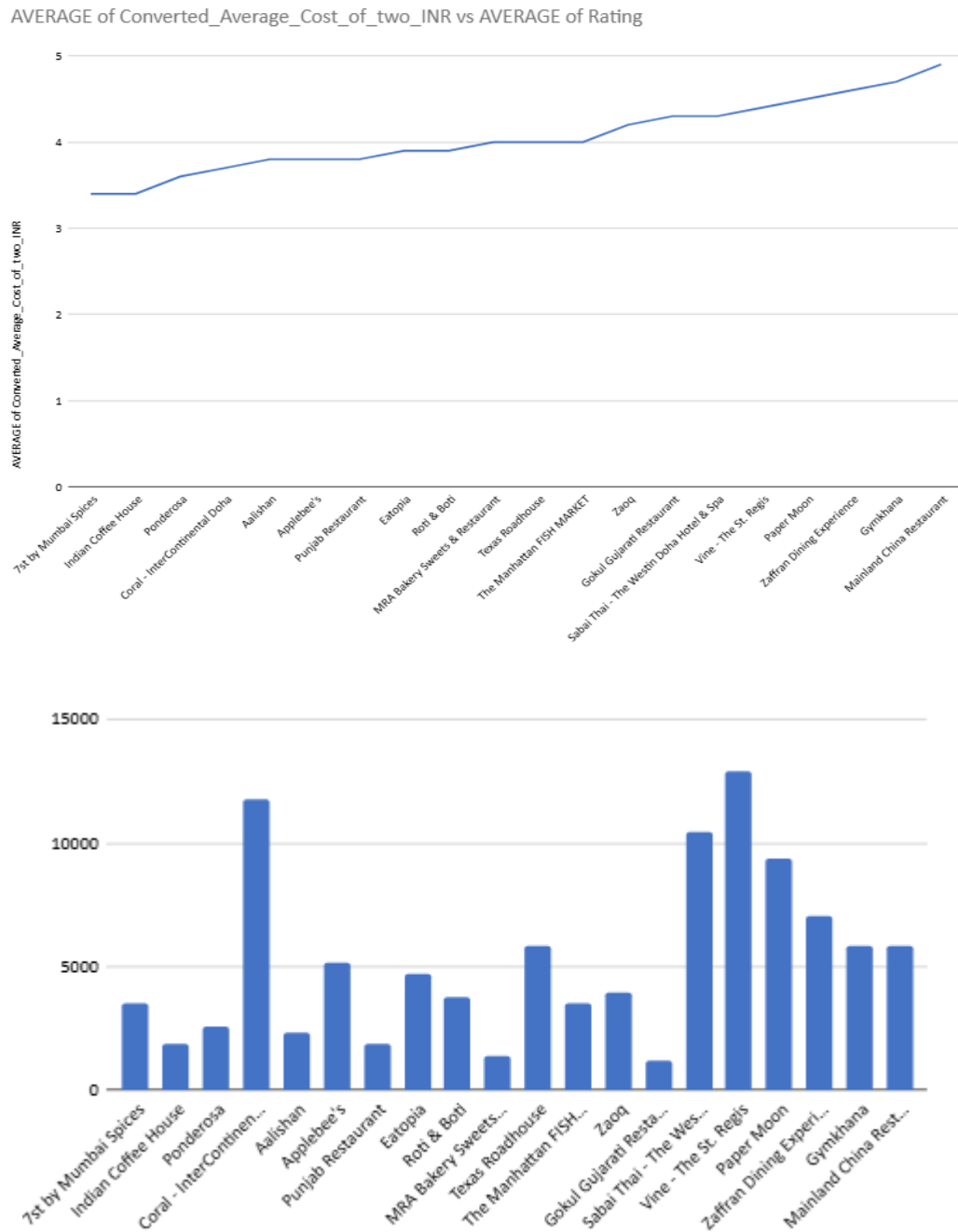
Step 1: I had utilized 3 pivot tables to understand the restaurant names, average ratings, price range and their average cost of two people in INR.

Step 2: For filtering competitors, I was sorting table in ascending order for low average ratings (1 - 2 and 2 - 3), selecting 1 - 2 price range and average cost of two people should be greater than or equal to country average cost of two people.

Step 3: We are analysing restaurants from the suggested countries like Qatar, New Zealand, Indonesia, Sri Lanka.

References:

RestaurantName	AVERAGE of Rati	AVERAGE of Con	AVERAGE of Pric
7st by Mumbai Spices	3.4	3522	4
Indian Coffee House	3.4	1878	3
Ponderosa	3.6	2583	3
Coral - InterContinental Doha	3.7	11740	4
Aalishan	3.8	2348	3
Applebee's	3.8	5166	4
Punjab Restaurant	3.8	1878	3
Eatopia	3.9	4696	4
Roti & Boti	3.9	3757	4
MRA Bakery Sweets & Restaur	4	1409	3
Texas Roadhouse	4	5870	4
The Manhattan FISH MARKET	4	3522	4
Zaoq	4.2	3992	4
Gokul Gujarati Restaurant	4.3	1174	2
Sabai Thai - The Westin Doha	4.3	10449	4
Vine - The St. Regis	4.4	12914	4
Paper Moon	4.5	9392	4
Zaffran Dining Experience	4.6	7044	4
Gymkhana	4.7	5870	4
Mainland China Restaurant	4.9	5870	4



Insights:

In Qatar (Doha):

7st by Mumbai Spices with rating 3.4, price range is 4 and Average cost of two is 3522.

Indian Coffee House with rating 3.4, price range is 3 and Average cost of two is 1878.

In New Zealand (Auckland and Wellington City):

In Auckland,

De Fontein Belgian Beer Café with rating 2.3, price range is 4 and average cost of two is 5171.

In Wellington City,

Wagamama with rating 3.7, price range is 4 and average cost of two is 3620.

In Indonesia (Bandung and Bogar):

Momo Milk with rating 3.7, price range 2 and average cost of two is 368.

In Sri Lanka (Colombo):

Elite Indian Restaurant with rating 2.4, price range is 2 and average cost of two is 513.

Queen's Café with rating 2.5, price range 3 and average cost of two is 570.

Suggestions:

In Qatar, we can focus on Quality-driven Indian mid-premium food.

In New Zealand, we can focus on Mid-premium with better experience.

In Indonesia, we can focus on Mid-tier quality and branding play.

In Sri Lanka, we can focus on Quality setup and maintain clean.

6. Which cuisines should we focus on in the newer restaurants to get better feedback?
Does the choice of cuisines affect the restaurant ratings?

Ans: For this, we can use some parameters, so we can get the cuisines easier.

Approach:

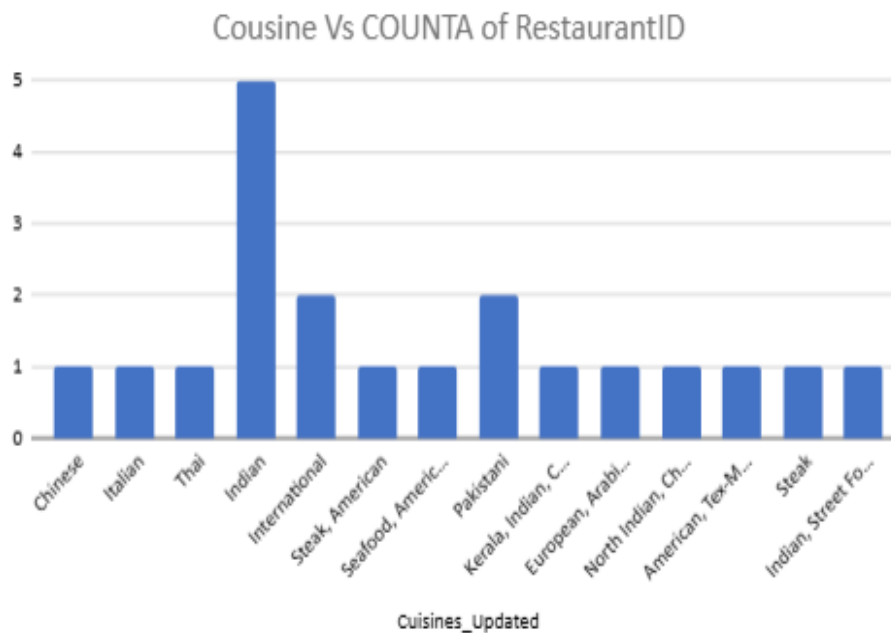
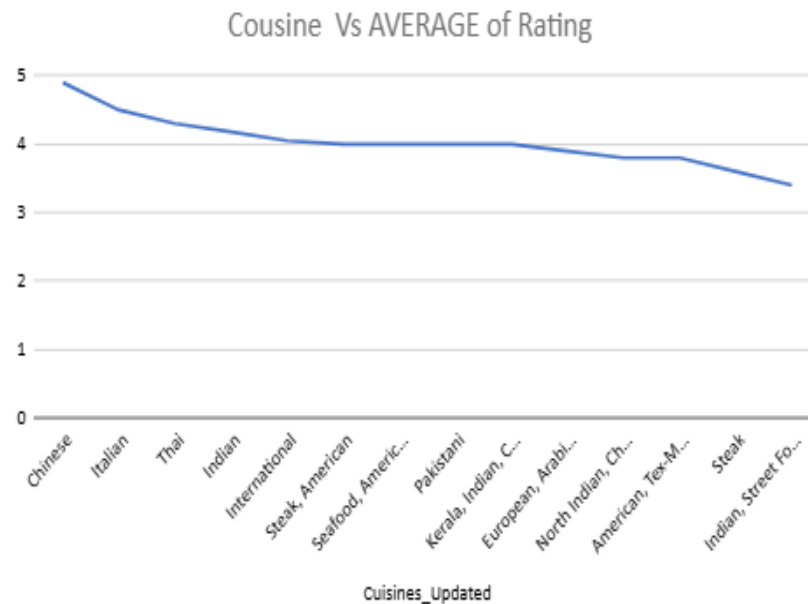
Step 1: For this I had a pivot table with cuisines as rows, count of restaurant IDs and average ratings as columns.

Step 2: For Getting the cuisines to get better feedback, I had sorted table in descending order based on average ratings.

Step 3: We are analysing cuisines from the suggested countries like Qatar, New Zealand, Indonesia, Sri Lanka.

References:

Cuisines_Updated	COUNTA of Rest	AVERAGE of Rati
Chinese	1	4.9
Italian	1	4.5
Thai	1	4.3
Indian	5	4.18
International	2	4.05
Steak, American	1	4
Seafood, American	1	4
Pakistani	2	4
Kerala, Indian, Chinese, Bakery	1	4
European, Arabian, Japanese, Bakery, Desserts	1	3.9
North Indian, Chinese, Turkish	1	3.8
American, Tex-Mex	1	3.8
Steak	1	3.6
Indian, Street Food	1	3.4



Insights:

High Performing Cuisines:

South Indian / North Indian / Indian / Seafood / Continental / Italian / Bakery / Brazilian / Sushi

These cuisines consistently achieved average ratings above 4.5, often ranging between 4.6 and 4.9, across multiple countries and formats.

Example:

- South Indian, Seafood, North Indian in Indonesia – Avg 4.9
- Bakery, Desserts in New Zealand – Avg 4.8
- Italian, North Indian, Continental in Sri Lanka – Avg 4.9
- Sushi in New Zealand – Avg 4.9
- Brazilian, Steak – Avg 4.7

Fusion formats like Italian + Pizza, Asian + Continental, and Indian + Asian also scored high minimal dilution, indicating appeal when curated carefully.

Lower Performing Cuisines:

Overloaded combinations such as North Indian + Chinese + Mughlai or Café + Bakery + Vegetarian + Tea saw low customer satisfaction.

Example:

- French, Japanese, Desserts – Avg 2.4 (Sri Lanka)
- Café, Bakery, Tea, Vegetarian- Avg 2.5
- North Indian, Fast Food, Italian, Asian – Avg 2.3 (New Zealand)

Standalone or overdone regional cuisines such as Turkish (3.4), Latin American + Italian (3.4), and Filipino, Japanese, Asian (3.7) underperformed.

Strategic Suggestions:

Cuisine	Average Rating	Why
Italian	High (>4.5)	Loved across all countries, especially in fusion formats
Indian (North & South)	Very High	Always a strong performer when paired smartly.
Continental	Consistently High	Appeals to a broad and diverse palate.
Bakery	High (when under 3 cuisines)	Performs well when focused and quality driven.
Sushi / Seafood	Premium Ratings	Considered authentic and fresh, Ideal for modern urban markets.
Brazilian/ Steak/ Bar Food	4.6 – 4.9 in multiple cases.	Great for upscale or nightlife-focused locations.

Avoid or Reassess Offering:

- North Indian, Chinese, Mughlai – Repetitive uniqueness to stand out.
- 3+ Cuisines in one menu – Lower Ratings across the board. Customers perceive this as scattered.
- Café, Tea, Vegetarian Mix – Works only in niche pockets. Generally, underperforms on scale.
- French + Japanese + Dessert – Lowest Rated Combo, lacks focus and likely fails to meet expectations in taste and theme.
- Turkish / Latin American (in small Clusters) – Poor Individual Performance unless highly localized or backed by authentic chefs.

Does Cuisine affect Ratings?

Yes, Cuisines strongly impact ratings due to customer expectations, regional acceptance, and quality control.

- **Global favourites** (Italian, Sushi, Bakery) and fresh, premium styles (Seafood, Brazilian) consistently performs well.
- **Regional favourites** (Indian, South Indian, Continental) also thrive when curated well.
- **Overloaded menus** or **Unfocused experimental fusions** lead to confused branding and lower satisfaction.

7. According to our current data, should we go for online delivery and table booking?
Does that affect the customer's ratings?

Ans: For this, we can use below 3 parameters, so we can understand better

(i) Has Table Booking: YES/NO

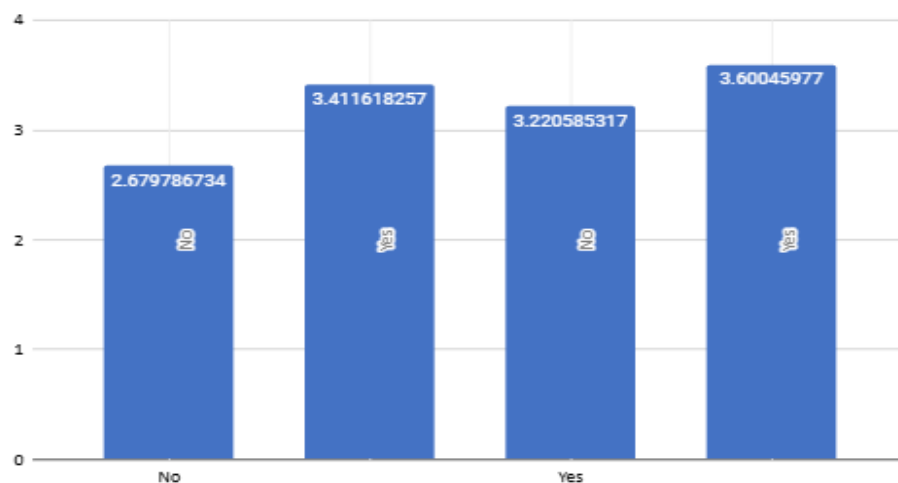
(ii) Has Online Booking: YES/NO

(iii) Average Ratings

With these parameters, we can make a pivot table and observe the values and decides what to do.

Has_Online_deliv	Has_Table_book	AVERAGE of Rat
No	No	2.679786734
	Yes	3.411618257
Yes	No	3.220585317
	Yes	3.60045977

References:



Insights:

Restaurants offers both Online delivery and table booking has a highest average rating of 3.60.

Restaurants offer Online delivery but not table booking had an average rating of 3.22, meanwhile restaurants with No Online delivery and table booking had an average rating of 3.41.

Restaurants with no Online delivery and no table booking had an average rating of 2.67.

Suggestions:

Adding Online delivery and table booking will improve the average ratings.

While, focusing only on table booking has a positive rating meanwhile, with only online delivery has moderate ratings.

yes, enabling online delivery and/or table booking tends to result in higher customer ratings, likely due to convenience and perceived service quality.

8. Should the team keep the rate of cuisines higher? Will that affect the feedback?

According to our data are the rates of cuisines and ratings, correlated?

Ans: For this, we can use below 3 parameters, so we can understand better

(i) Cuisine

(ii) Converted average cost of two

(iii) Average ratings

With these parameters we can draw the pivot table and to obtain the relationship between two numerical columns.

Draw the scatter plot between Converted average cost of two and ratings using Raw_Data.

Formula:

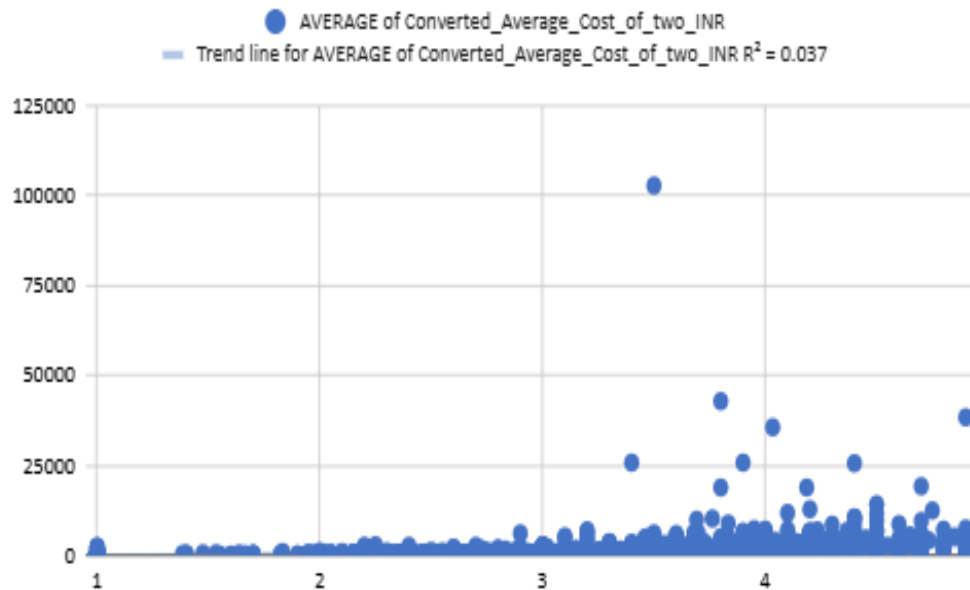
=CORREL('Raw Data'!Y2:Y9552, 'Raw Data'!AA2:AA9552)

= 0.1484

References:

Cuisines_Updated	AVERAGE of Con	AVERAGE of Rat
Afghani	513	1.475
Afghani, Mughlai, Chinese	500	1
Afghani, North Indian	900	1
Afghani, North Indian, Pakistani, Arabian	500	1
African	2152	4.7
African, Portuguese	3725	4.2
American	10193	3.7625
American, Asian, Burger	3419	4.6
American, Asian, European, Seafood	2000	4.6
American, Asian, Italian, Seafood	11514	4.5
American, Asian, North Indian, Mexican, South Indian	500	3.3
American, Bakery, European, Burger, Fusion	4274	3.8
American, Bar Food	1817	4.225
American, Bar Food, BBQ	2137	1
American, Bar Food, Sandwich	2137	3.7
American, BBQ	1496	3.8
American, BBQ, Burger	2137	3.85
American, BBQ, Sandwich	2992	4.9
American, BBQ, Seafood	2137	3.3
American, BBQ, Southern	1496	3.95
American, BBQ, Steak	2565	3.825
American, Breakfast	1832	3.814285714
American, Breakfast, Burger	1710	4.033333333
American, Breakfast, Desserts	855	4.3
American, Breakfast, Diner	855	3.4
American, Breakfast, Greek	855	4
American, Breakfast, Hawaiian	2137	4.2
American, Breakfast, Vegetarian	2137	4.4
American, British, Continental	1700	4
American, Burger	2583	4.076923077
American, Burger, Cafe	800	4.7
American, Burger, Cajun	2137	3.8
American, Burger, Fast Food	500	4.6
American, Burger, Grill	5870	4.9

AVERAGE of Converted_Average_Cost_of_two_INR and AVERAGE of Rating



Insights:

Corelation between Converted average cost of two and ratings is close to 0.15.

This means, prices do not significantly impact the customer feedback.

It also says One more thing, customers give the ratings based on quality, service and experience.

Suggestions:

Don't raise the prices just to appear as premium it doesn't affect the feedback

Instead focus on Quality, Service etc.

9. What is the distribution of the number of restaurants of different price ranges in all the countries?

Ans: From the below pivot table,

Price range 1 contains a total of 4444 (Cheap)

Price range 2 contains a total of 3113 (Moderate)

Price range 3 contains a total of 1408 (Premium)

Price range 4 contains a total of 584 (Luxury)

Price_range	COUNTA of Rest
1	4444
2	3113
3	1408
4	586
Grand Total	9551

Below pivot table, gives the different price ranges country wise

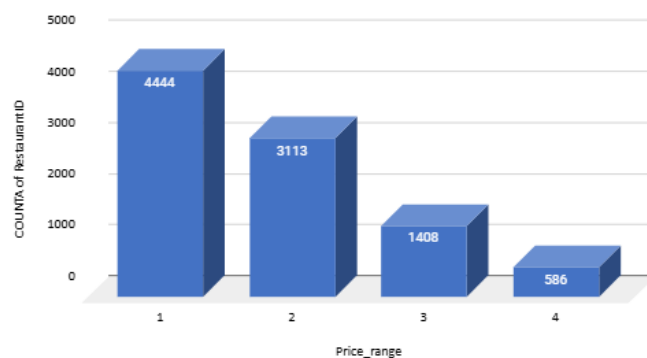
COUNTA of Resti	Price_range				
Country Name	1	2	3	4	Grand Total
Australia	4	14	5	1	24
Brazil	2	7	16	35	60
Canada		3		1	4
India	4295	2858	1111	388	8652
Indonesia		1	20		21
New Zealand	3	4	17	16	40
Philippines		1	12	9	22
Qatar		1	5	14	20
Singapore		1	5	14	20
South Africa		4	17	39	60
Sri Lanka		6	11	3	20
Turkey		11	18	5	34
United Arab Emirates		9	29	22	60
United Kingdom	4	28	32	16	80
United States of	136	165	110	23	434
Grand Total	4444	3113	1408	586	9551

This will help us to identify where price range 1 are dominant and where price range 4 is most common.

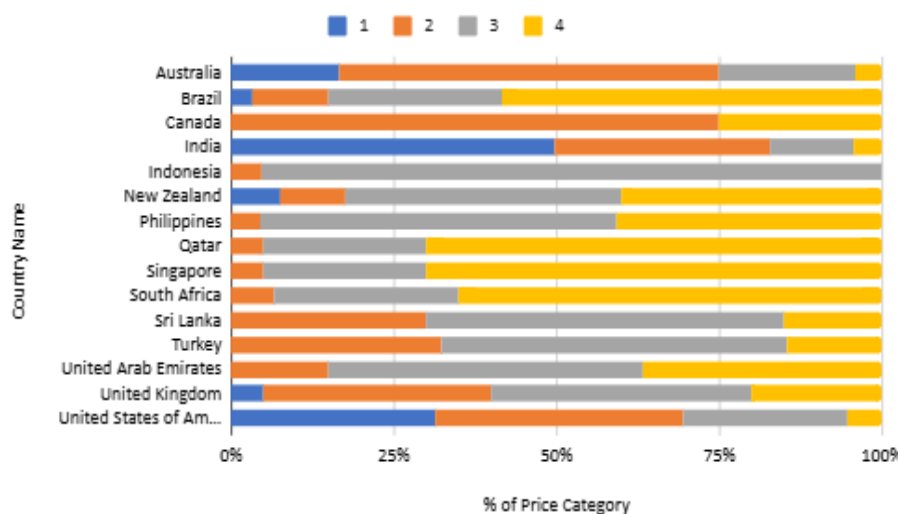
We can visualize it by stacked column chart.

References:

COUNTA of RestaurantID vs Price_range



Country Wise Count of Restaurants per Price Category



Insights:

Price range 1, had largest number of restaurants, which means they mainly focus on budget minded customers.

Price range 2, had similar number of customers, which means they are mid-range dining mindset customers.

Price range 1 & 2 are very competitive

Price range 3 & 4, had fewer restaurants, much room for expansion.

Suggestions:

Avoid only focusing on Price range 1.

Explore Price Range 2 with better quality.

Make Price Range 3 & 4 as Premium Service and Brand Flagships restaurants.

10. Explain your approach in brief for suggesting countries/cities in order to open new restaurants, if the objective and subjective questions would have not been given to assist you.

Ans: Here is my approach,

- Initially, I will identify the areas which has online delivery, Table Booking and Is_delivering_now using geo clusters using latitude and longitude columns.
- In that, we will identify the areas where customers are under served, so we may have high chances.
- For, identifying the cities, I will get the % of restaurants in that city, where they have no online delivery, no table booking and not delivering currently.
- Second, I will use the votes column to my favour. Here higher votes indicate that customers are actively voting which also means that customers are actively looking for new restaurants.
- It suggests, they are the good markets.
- Third, for financial budget calculation, I will use currency conversions and average cost of two as usual, and understand the affordability and purchasing power.
- Sort the countries, where cost is manageable and currency is stable
- Combine this with price range, and see whether the people can pay premium or economy prices.
- Fourth, for selecting the cuisines count the frequencies of each cuisine in the city.
- Identify the cities with 3-4 cuisines repeated, which means they are less diverse and are willing to try new restaurants. We can add them.

- For following upto date trends, obtain year from Datekey_Opening, so that we know which city/country has a rise in opening restaurants recently.
- We know that newer restaurants follow new trends. So, there will be a few big competitors.
- For identifying big competitors, we follow few steps here:
 - i. Shortlist restaurants from recommended countries/cities with high average rating and high number of votes.
 - ii. Also include restaurants with online delivery and table booking and check whether they are in premium price range.
 - iii. And which is their top performing cuisine.